Tasks: Object detection and recognition

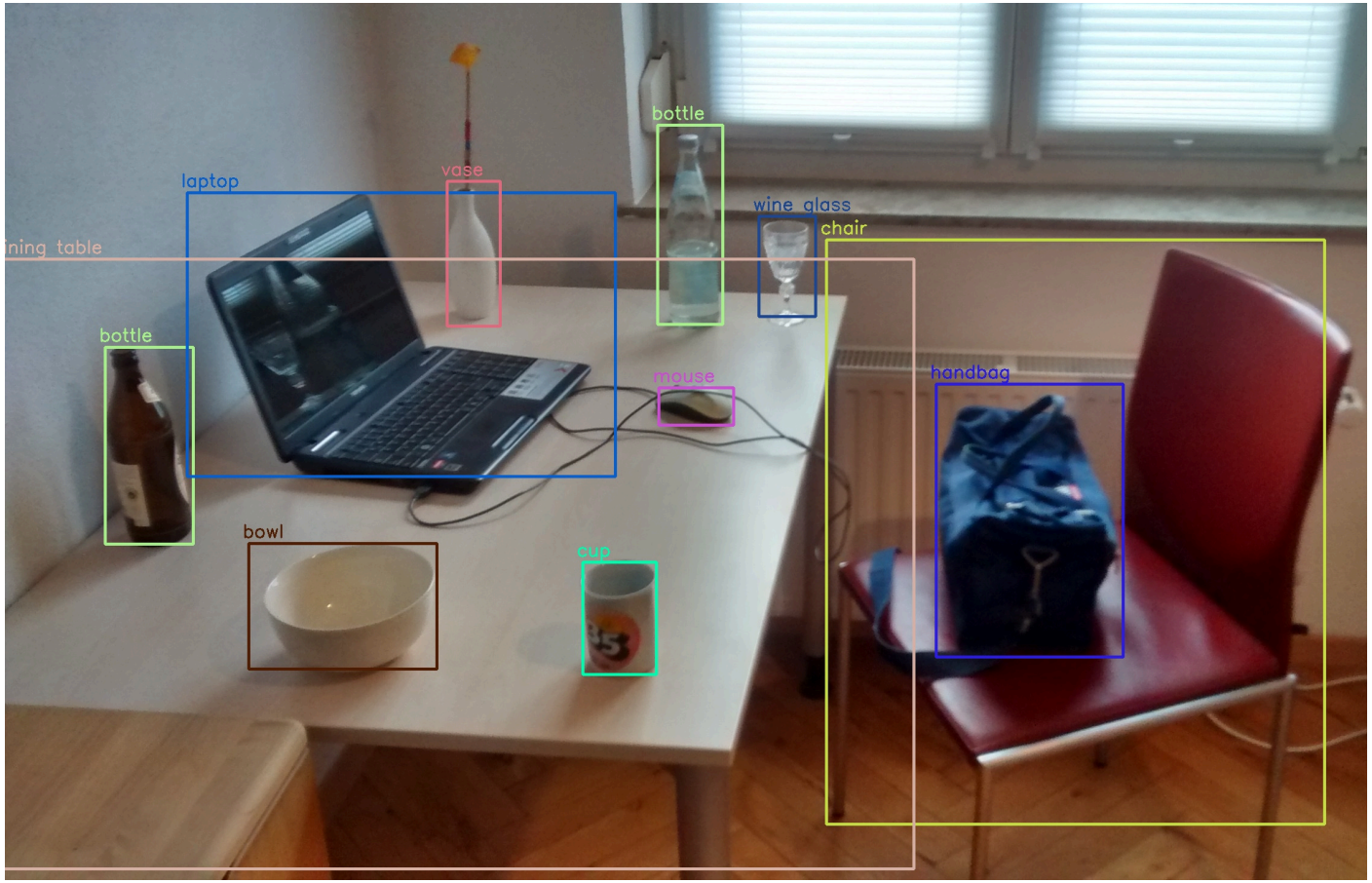


Image source: MTheiler, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Detected-with-YOLO--Schreibtisch-mit-Objekten.jpg

Tasks: Image segmentation

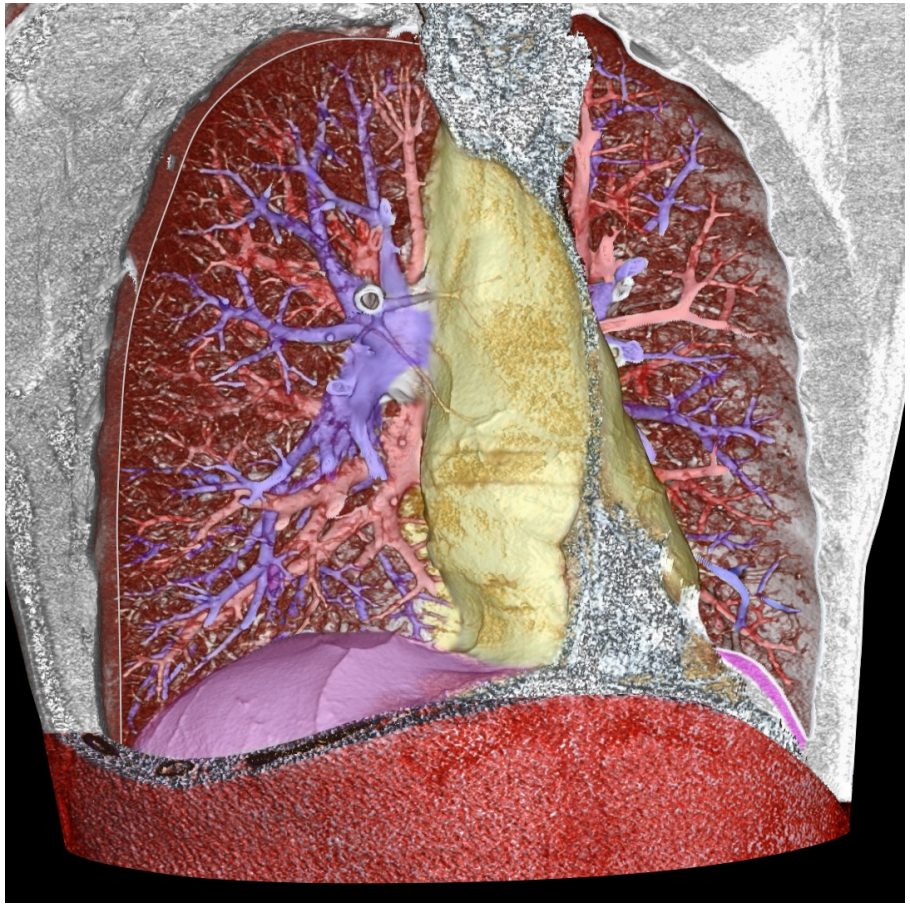


Image source: Mikael Häggström, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:3D_CT_of_thorax.jpg

Tasks: 3D reconstruction

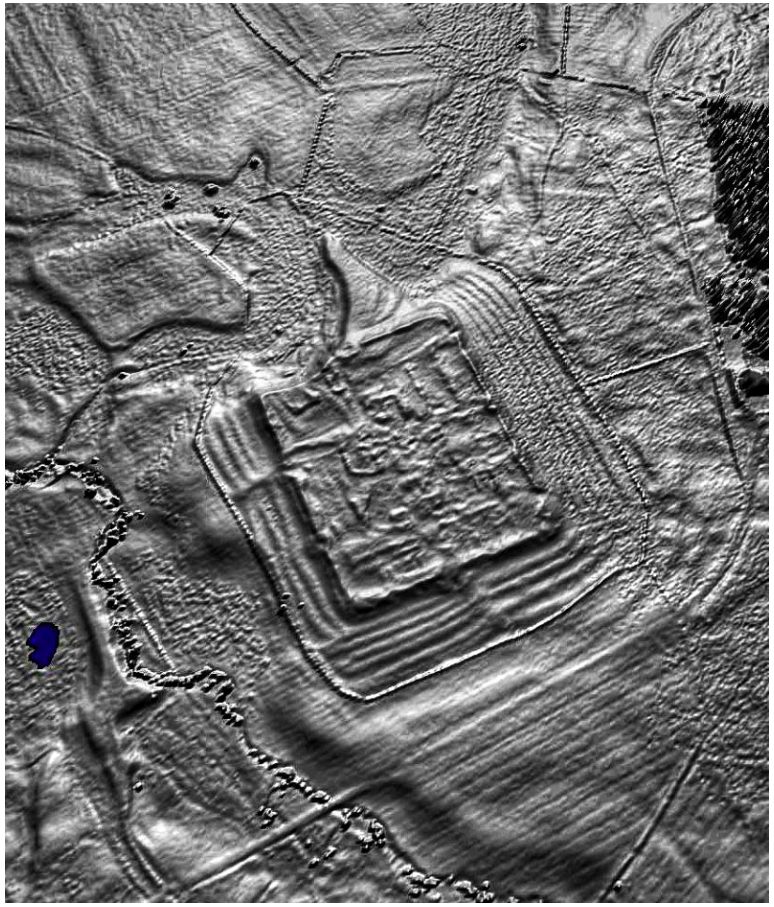


Image source: Dr John Wells, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:A_lidar_view_of_Habitancum_Roman_fort.jpg

Tasks: Pose estimation



Image source: Bing Image Creator, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:DALL-E2_-_Analogue_film_taken_with_Nikon_F2_showing_anthropomorphic_rhinoceros_wearing_a_business_suit_doing_some_cool_breakdance_moves_inside_indoor_ballet_room_studio,_pose_estimation_imposed_on_him.jpg

Tasks: Optical character recognition

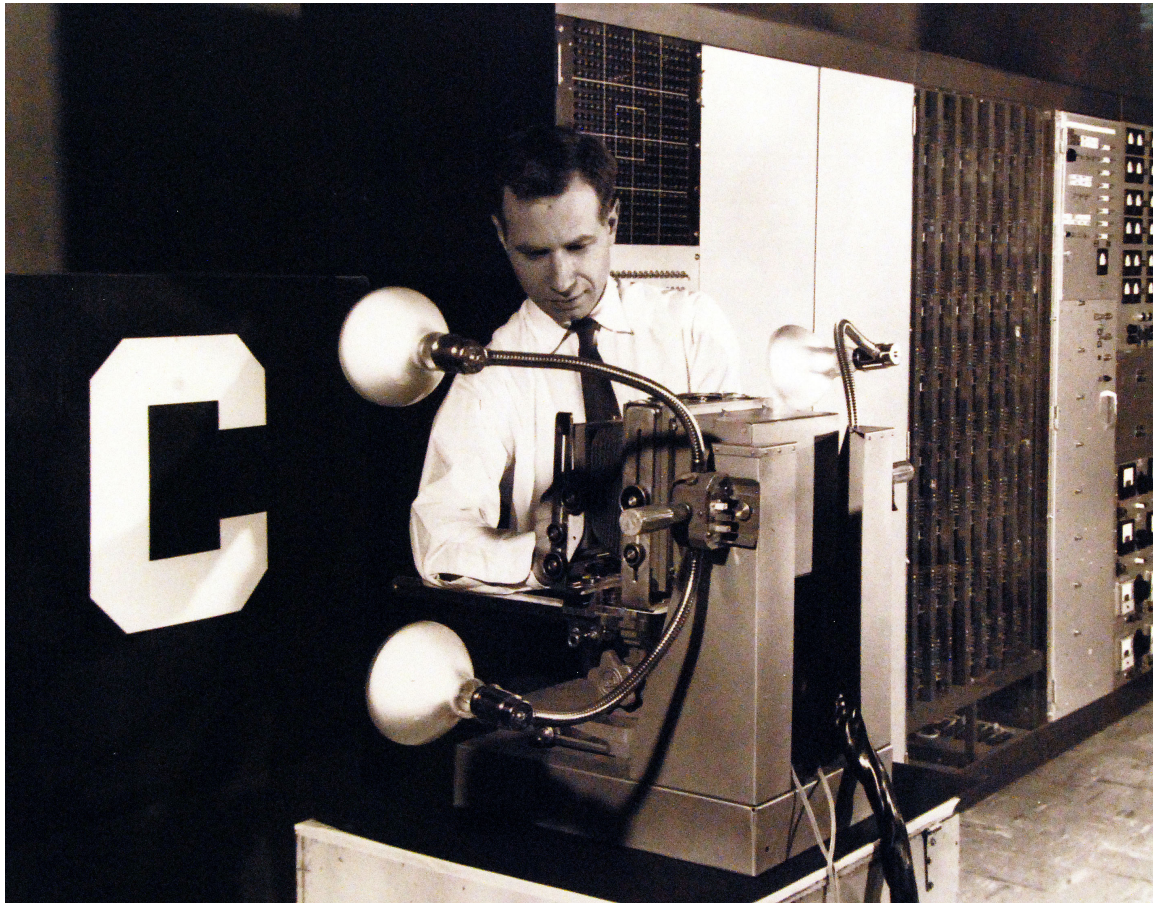


Image source: National Museum of the U.S. Navy, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:330-PSA-80-60_(USN_710739)_(20897323365).jpg

Tasks: Face recognition

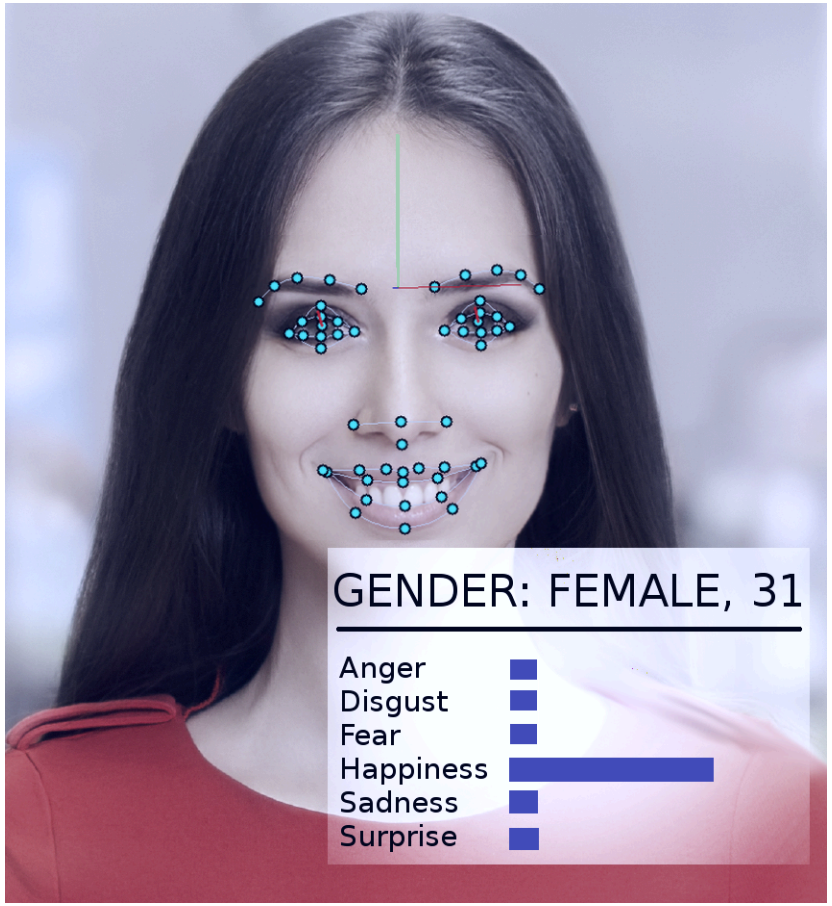


Image source: Abyssus, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Visage_Technologies_Face_Tracking_and_Analysis.png

Tasks: Egomotion

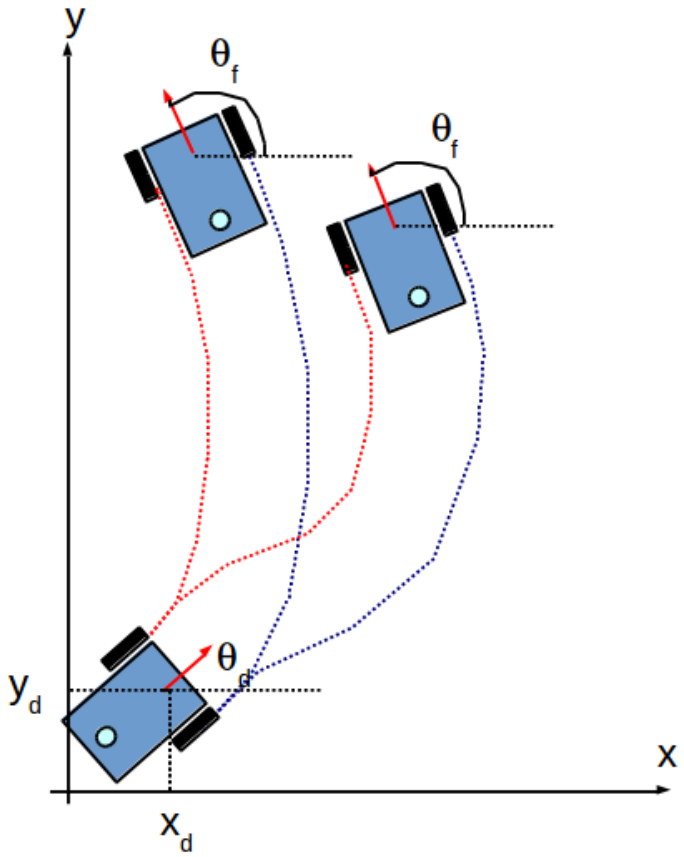


Image source: SergeMoutou, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:MobileRobotOdometry.png

Tasks: Optical flow

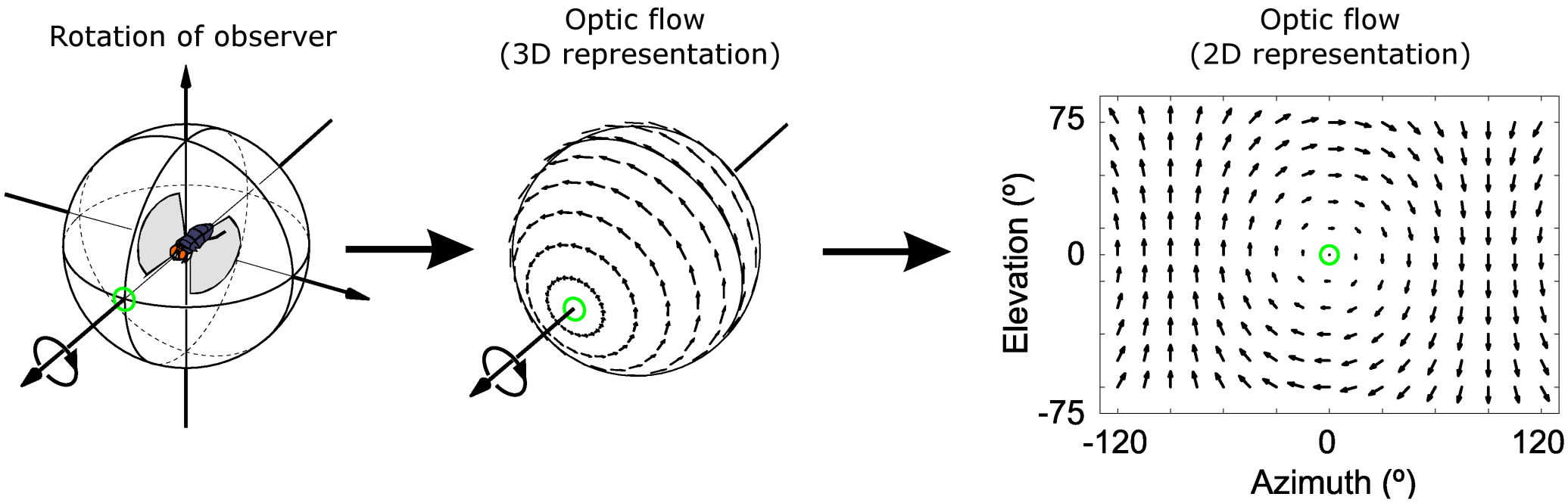


Image source: Adapted from PloS Biology article: Huston SJ, Krapp HG, 2008 Visuomotor Transformation in the Fly Gaze Stabilization System, CC BY 2.5 (https://creativecommons.org/licenses/by/2.5), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Opticfloweg.png

What is Machine Learning?

From Wikipedia, page "Machine learning":

> Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data, and thus perform tasks without explicit instructions. Within a subdiscipline in machine learning, advances in the field of deep learning have allowed neural networks, a class of statistical algorithms, to surpass many previous machine learning approaches in performance.

What is Machine Learning?

Machine learning is a broad topic that covers:

- Computer vision
- Natural language processing
- Deep learning
- Reinforcement learning

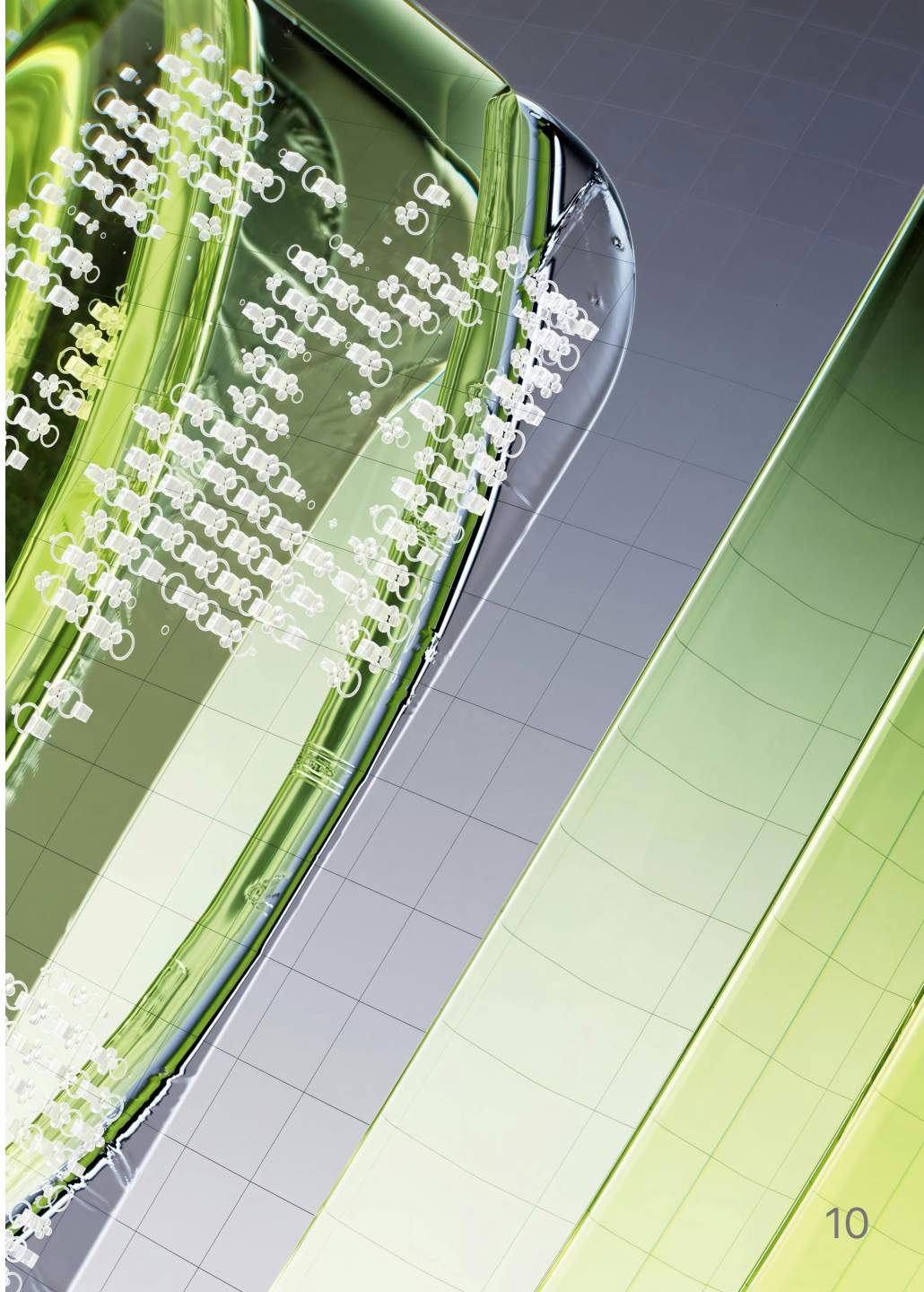


Image source: https://www.pexels.com/photo/an-ai-generated-graphic-25630347/

Tasks: Classification

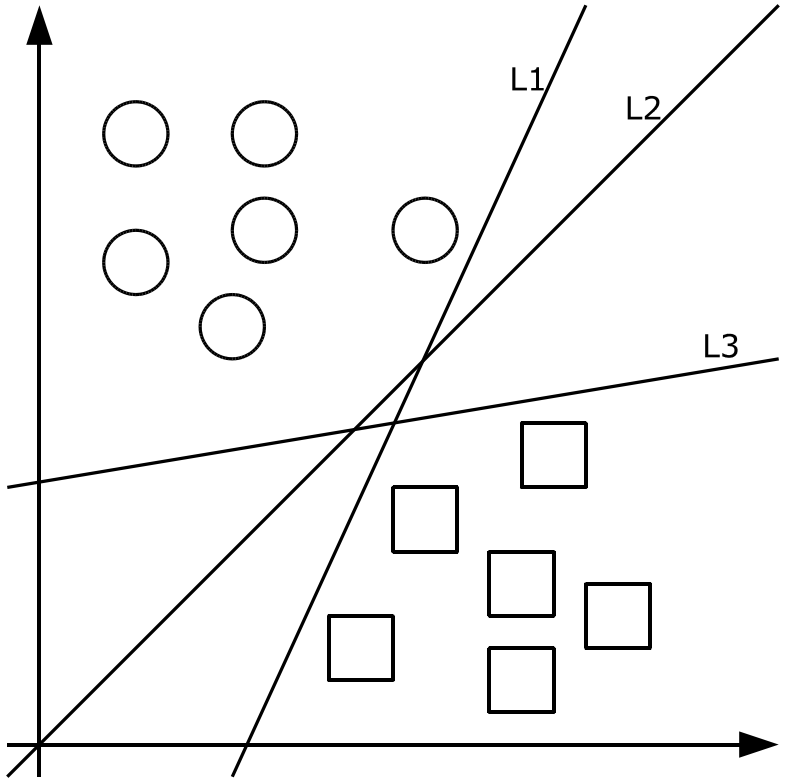


Image source: DJGulp3, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Classifier.svg

Tasks: Regression

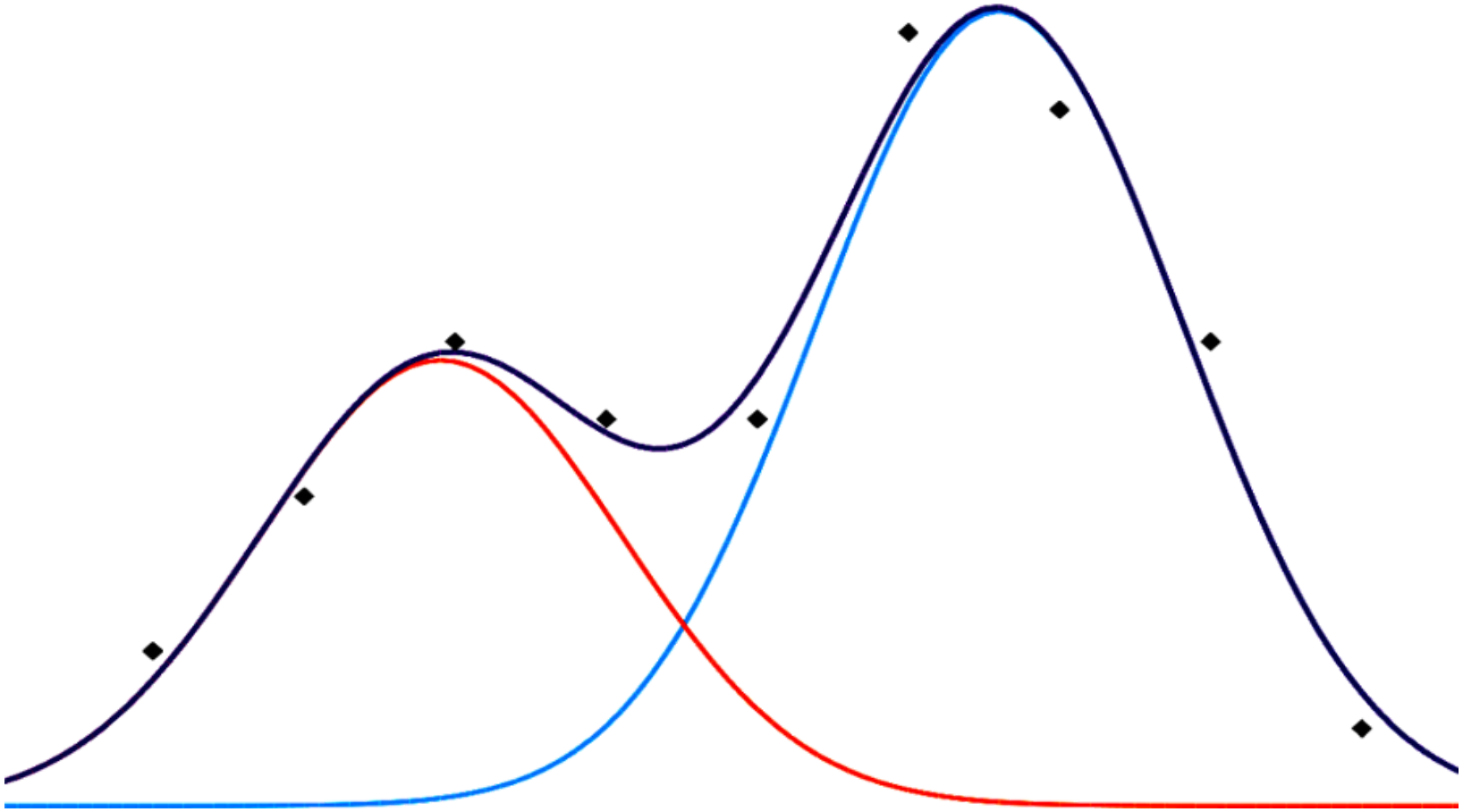


Image source: Mathieu ROUAUD, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:RegressionNonLineaireGaussiennes.png

Tasks: Dimensionality Reduction

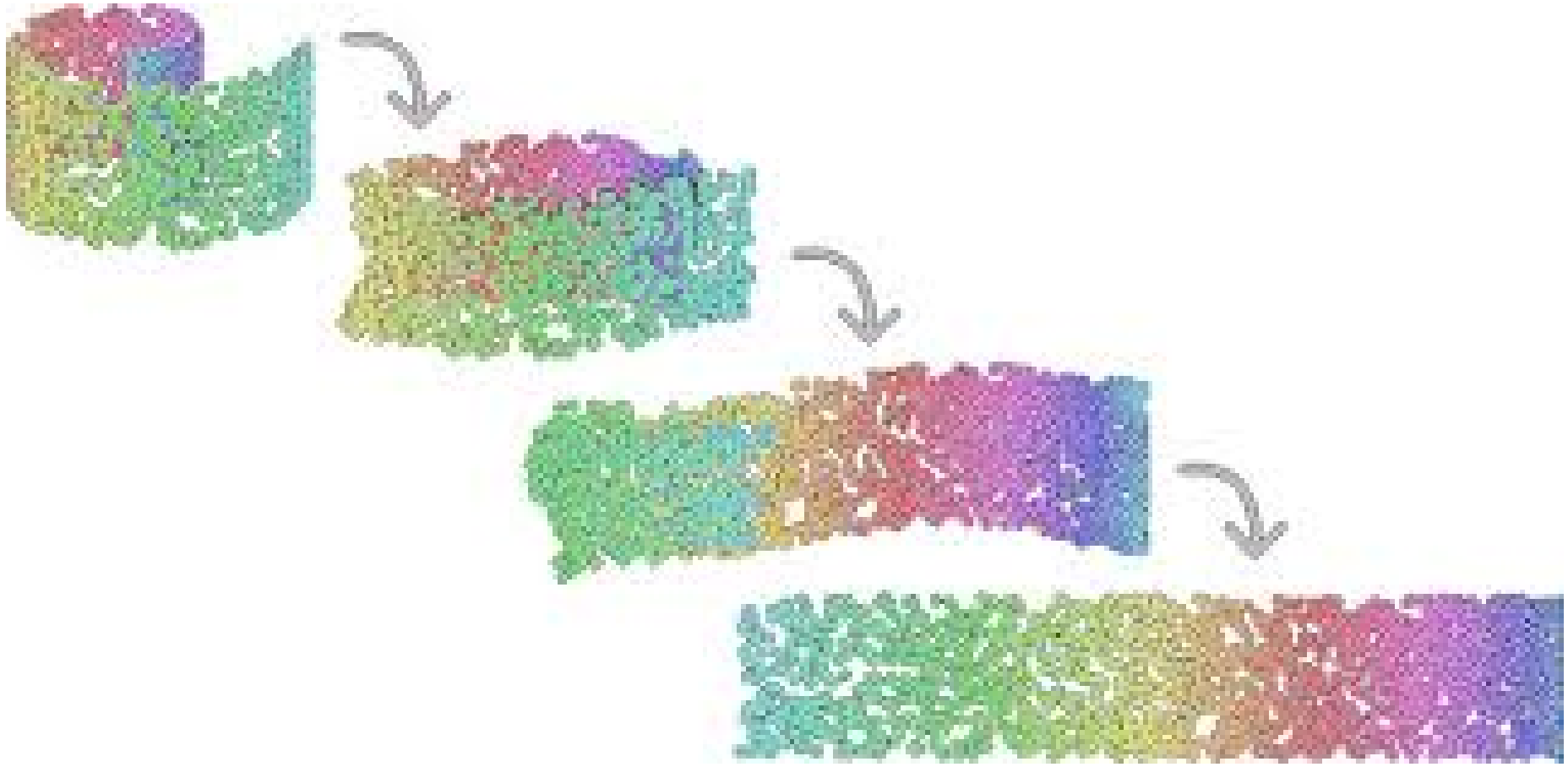


Image source: Michael Gashler, Dan Ventura, Tony Martinez, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Dimensionality-reduction.jpg

Tasks: Reinforcement Learning

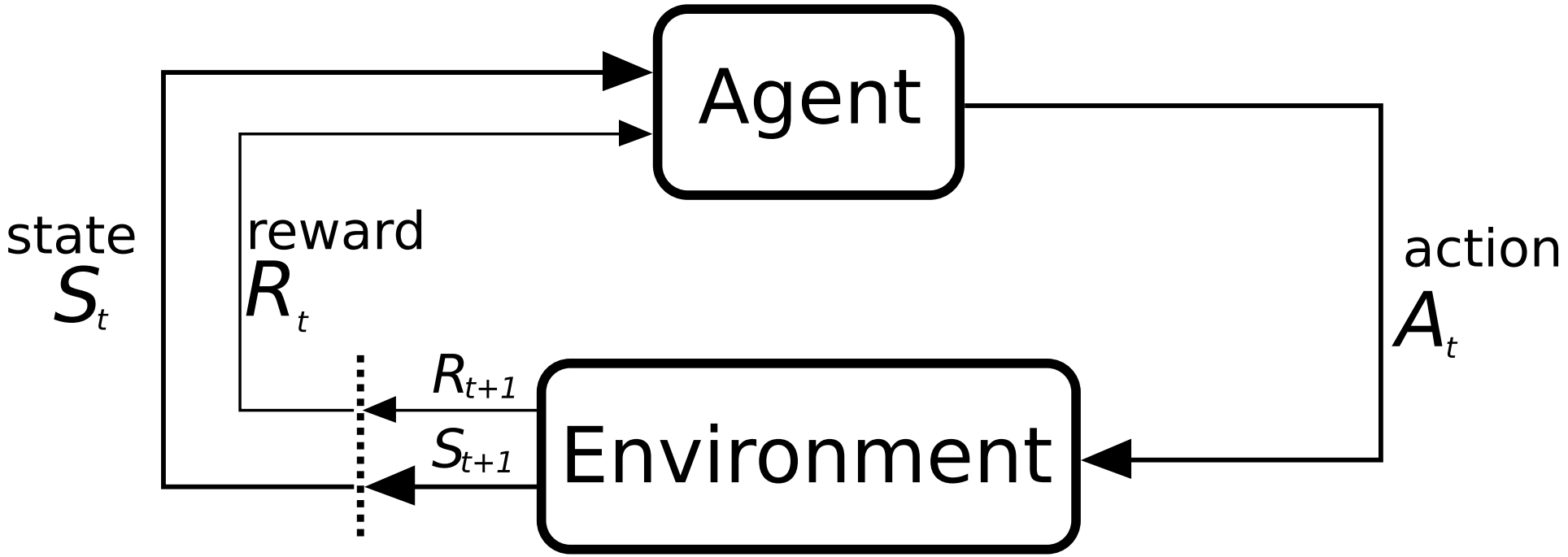


Image source: EBatlleP, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Markov_diagram_v2.svg

What is Natural Language Processing?

From Wikipedia, page "Natural language processing":

> Natural language processing (NLP) is the processing of natural language information by a computer. NLP is a subfield of computer science and is closely associated with artificial intelligence.

Why is Natural Language Processing important?

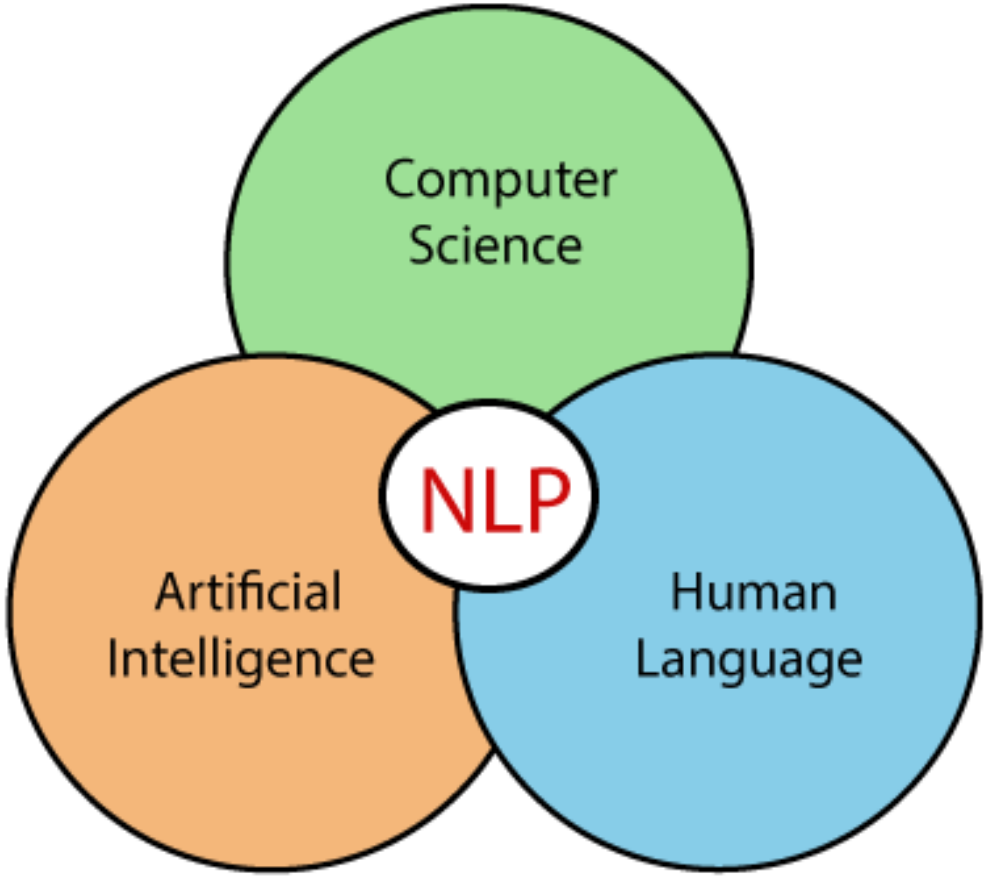
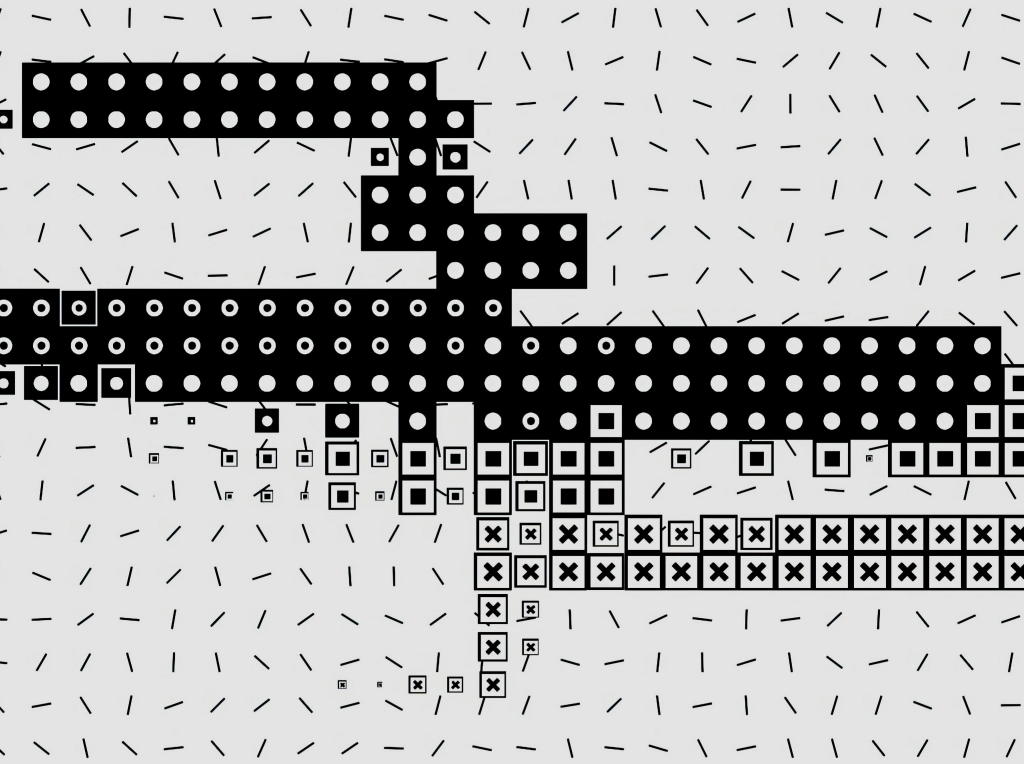


Image source: javatpoint, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:What-is-nlp.png



Why is Natural Language Processing important?

Natural language is, in my opinion, an important factor in modern robotics. Why?

- Natural language allows easy and direct communication between humans and robots.
- Natural language allows to access existing knowledge, e.g. documentation, for use in robotics.

Image source: https://www.pexels.com/photo/symbolic-graphic-representation-25626431/

Yes, this is a Deep Learning course!

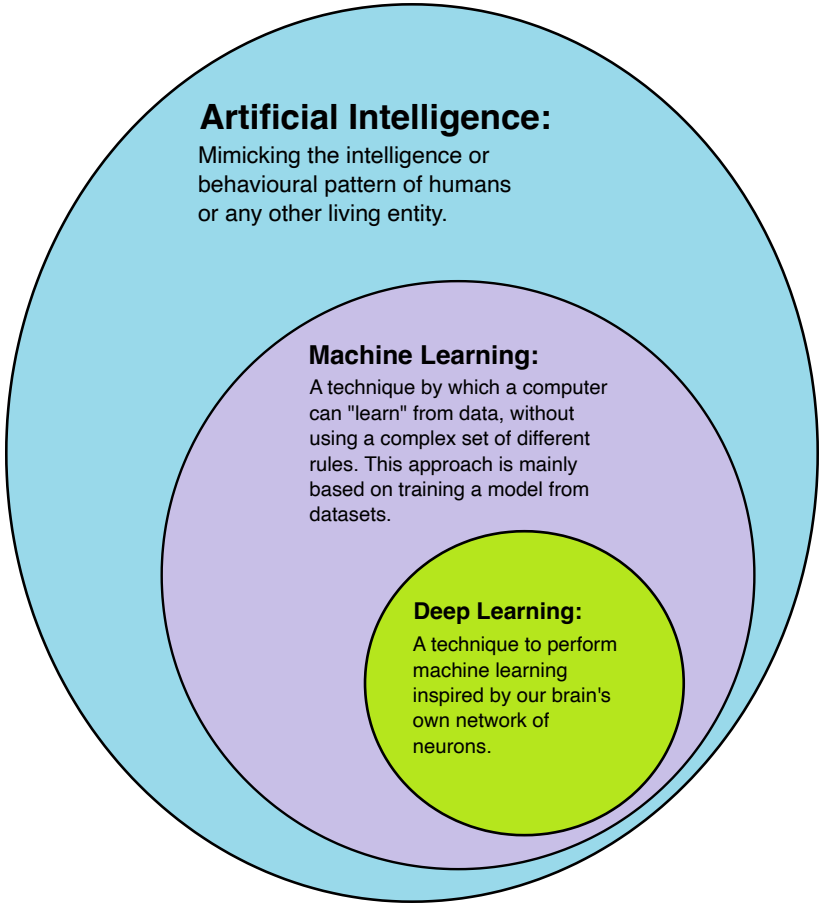


Image source: Avimanyu786, Tukijaaliwa, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:AI-ML-DL.svg

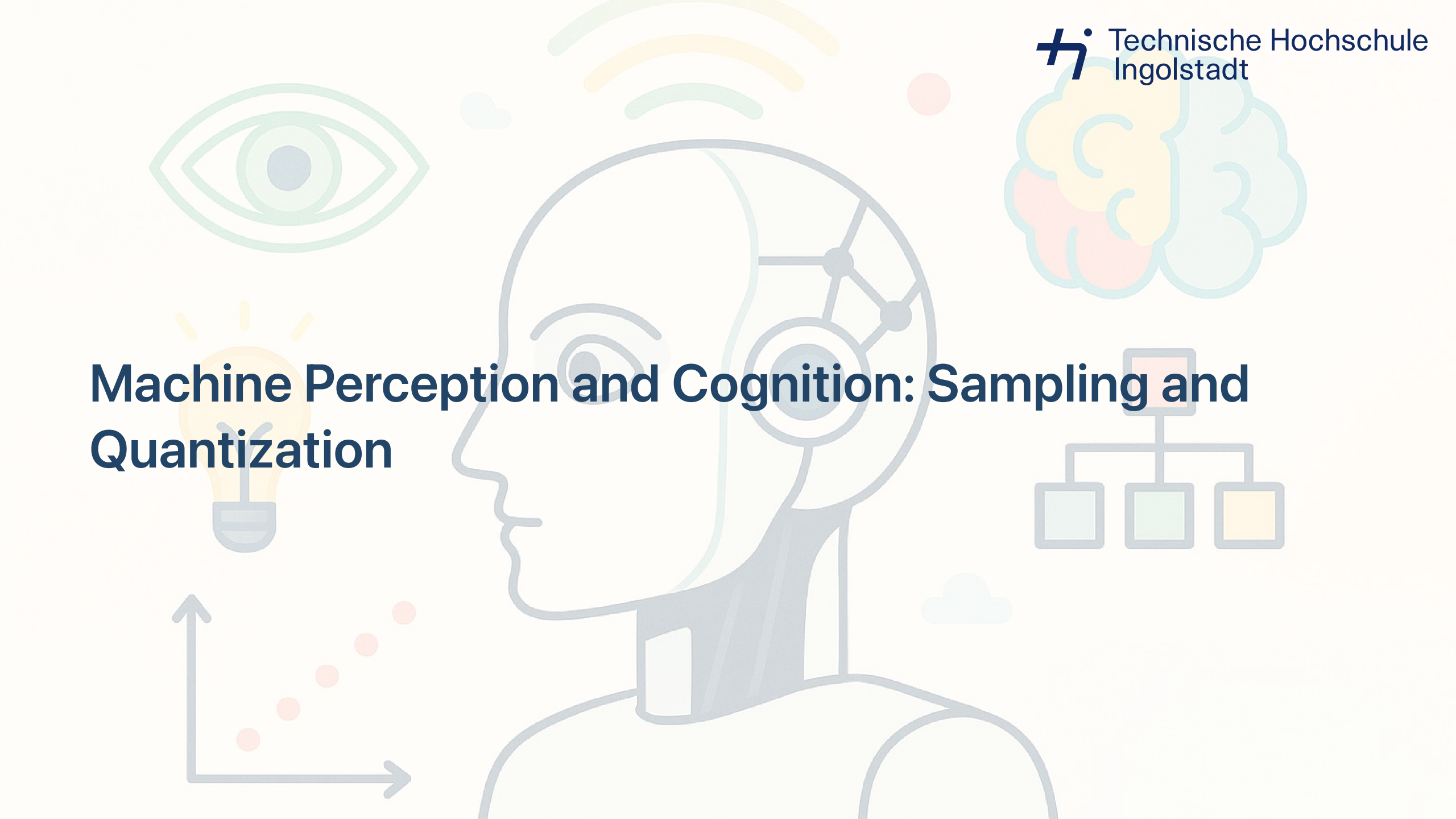
What will we do in this course? An overview...

- Physical to digital signals.
- Sensor types and their properties.
- Artificial neural networks and training them.
- Convolutional and recurrent neural networks and their uses.

What will we do in this course? An overview...

- Handwriting and speech recognition.
- Attention networks and transformers.
- Large language models and their uses.
- Vision-language and vision-language-action models and their uses.

Technische Hochschule Ingolstadt



Machine Perception and Cognition: Sampling and Quantization

What is the problem?

Physical signals have the following properties:

- Continuous over time.
- Continuous over amplitude.
- Think of "infinite detail".

What is the problem?

You can zoom in on any point and still find more detail!

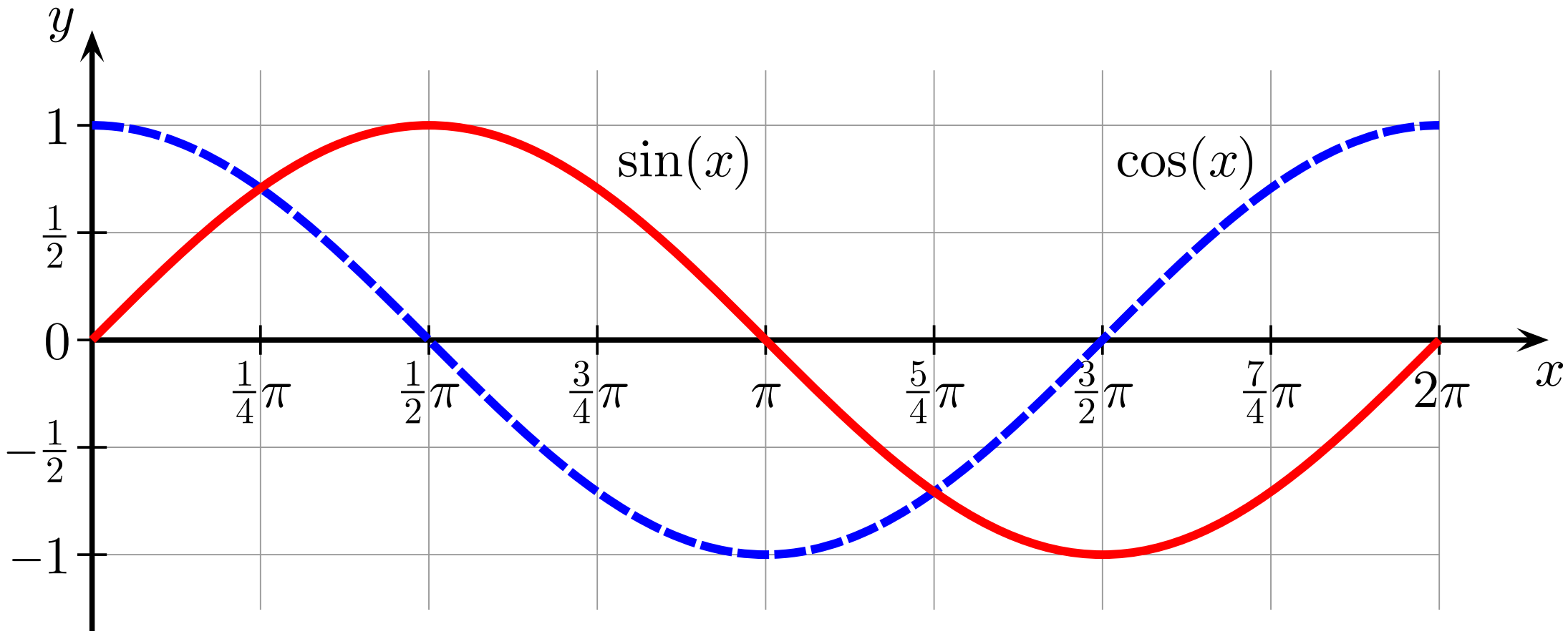


Image source: Geek3, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Sine_cosine_one_period.svg

Fun fact!

The graphic is in SVG (scalable vector graphics), which allows to mathematically describe shapes instead of storing a rasterized image. It can therefore actually zoom and reveal more detail!

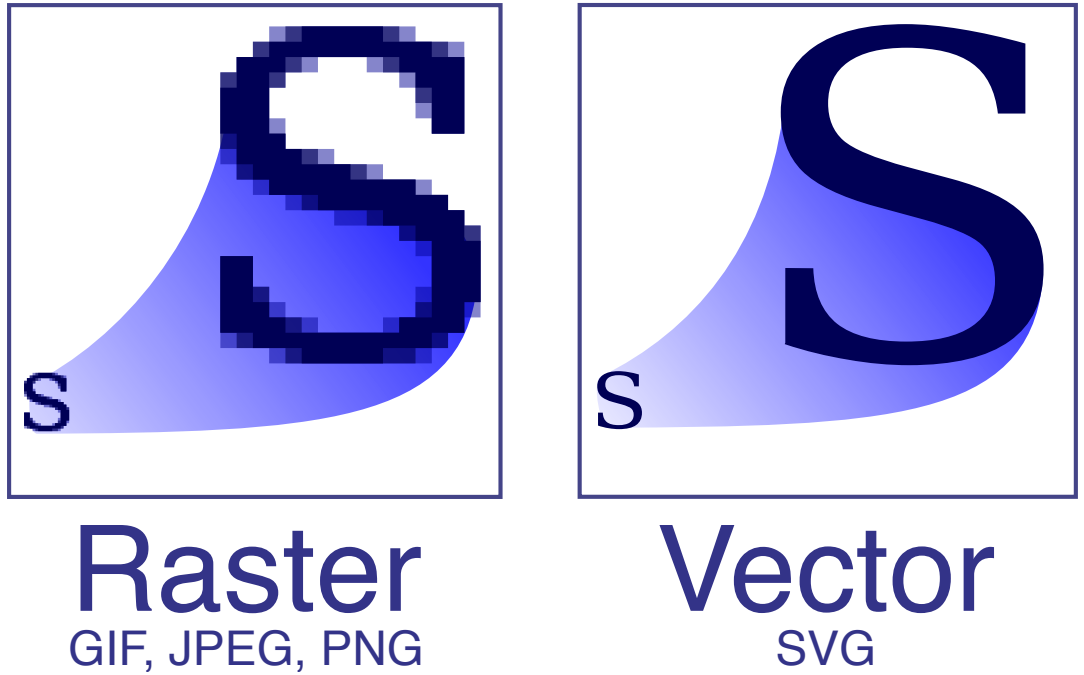
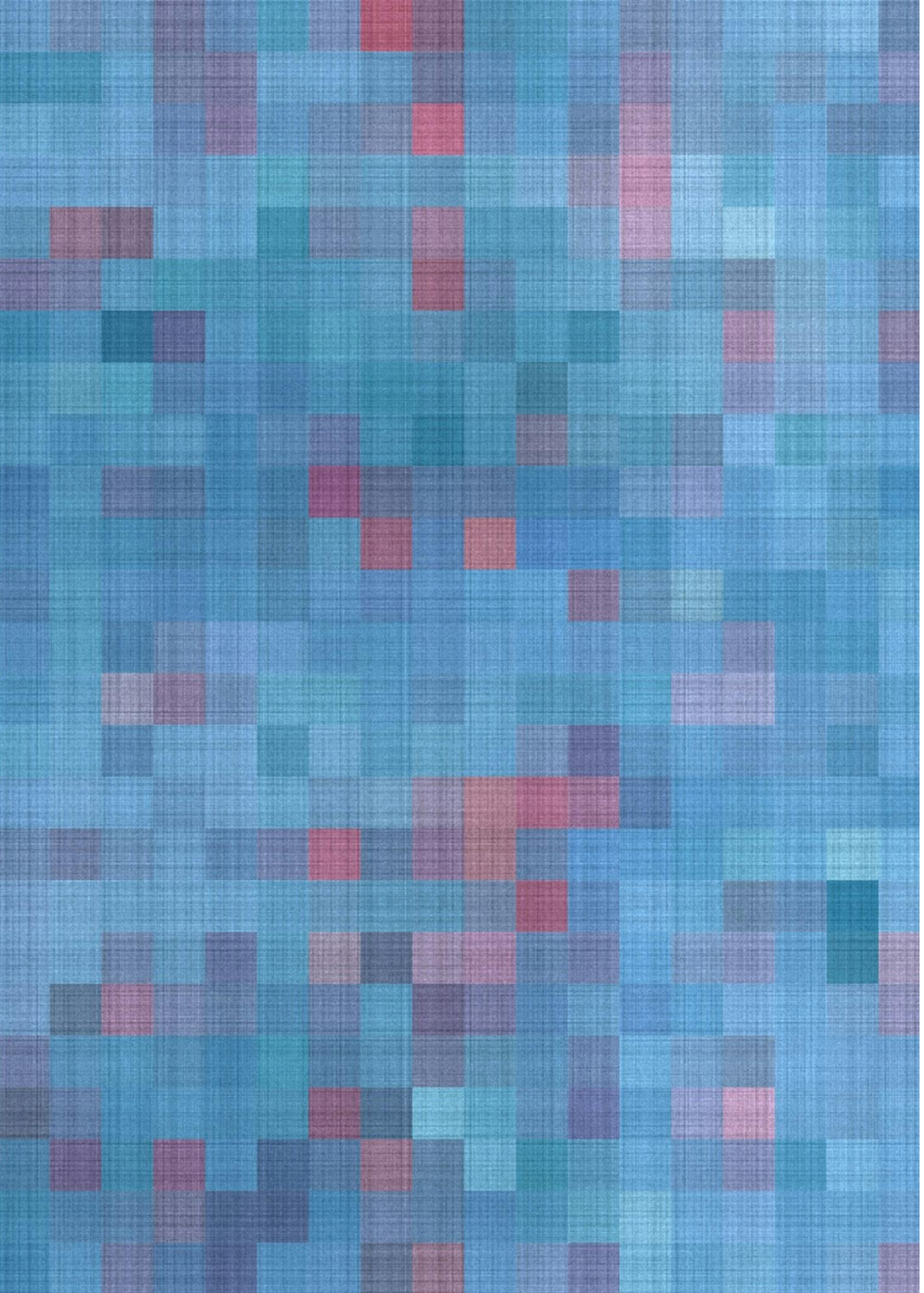


Image source: Yug, modifications by Cfaerber et al., CC BY-SA 2.5 (https://creativecommons.org/licenses/by/2.5), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Bitmap_VS_SVG.svg

What is the problem?

On the other hand, computers have the following properties:

- Limited memory (volatile and persistent).
- Limited processing power.
- Limited representation of values.



What is the problem?

Sensors also have limited resolution. Think of the number of pixels per row and column in a camera image.

Image source: https://www.pexels.com/photo/close-up-photo-of-textile-7026427/

Sampling and quantization

Sampling: Reduction of a time-/space-continuous signal to a time-/space-discrete samples.
Quantization: Reduction of an amplitude-continuous signal to discrete values.

... or ...

Sampling: Discretization in time and/or space.
Quantization: Discretization in amplitude.

Sampling

From Wikipedia, page "Sampling (Signal processing)":

> In signal processing, sampling is the reduction of a continuous-time signal to a discrete-time signal. A common example is the conversion of a sound wave to a sequence of "samples". A sample is a value of the signal at a point in time and/or space; this definition differs from the term's usage in statistics, which refers to a set of such values.

Sampling

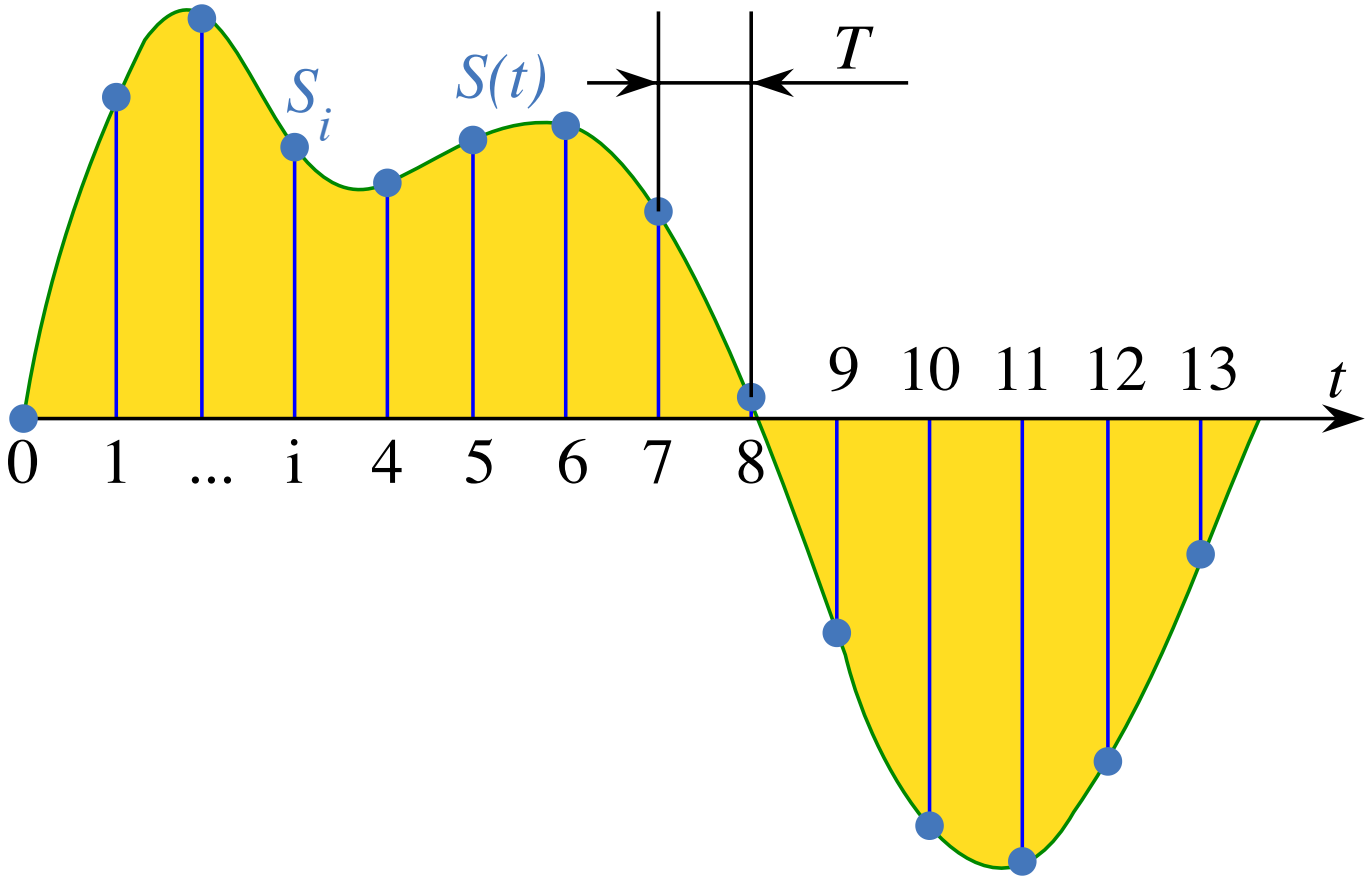


Image source: Д.Ильин, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Signal_Sampling.svg

Sampling

- Continuous signal s at time t written as $s(t)$
- In this course we restrict ourselves to uniform sampling:
  - Thus, sampling is done by measuring the signal $s(t)$ at evenly spaced intervals.
- Time interval is given in T seconds with $t_{i+1} - t_i = T$

Sampling

- Sampling frequency f_s is defined as $f_s = \frac{1}{T}$
- Sampling frequency is in units of *Hertz* (Hz), that is samples per second.

Sampling

An example:

- Sampling frequency $f_s = 100$ in Hertz.
- Sampling interval $T = \frac{1}{f_s} = \frac{1}{100} = 0.01$ in seconds.

Sampling in space

Sampling does not have to be over time only. It can also be over space, multi-dimensional space or a combination of space and time.

Think of e.g. a rasterized image!

We will later discuss potential problems with sampling and this applies to e.g. rasterized images as well.

Sampling in space

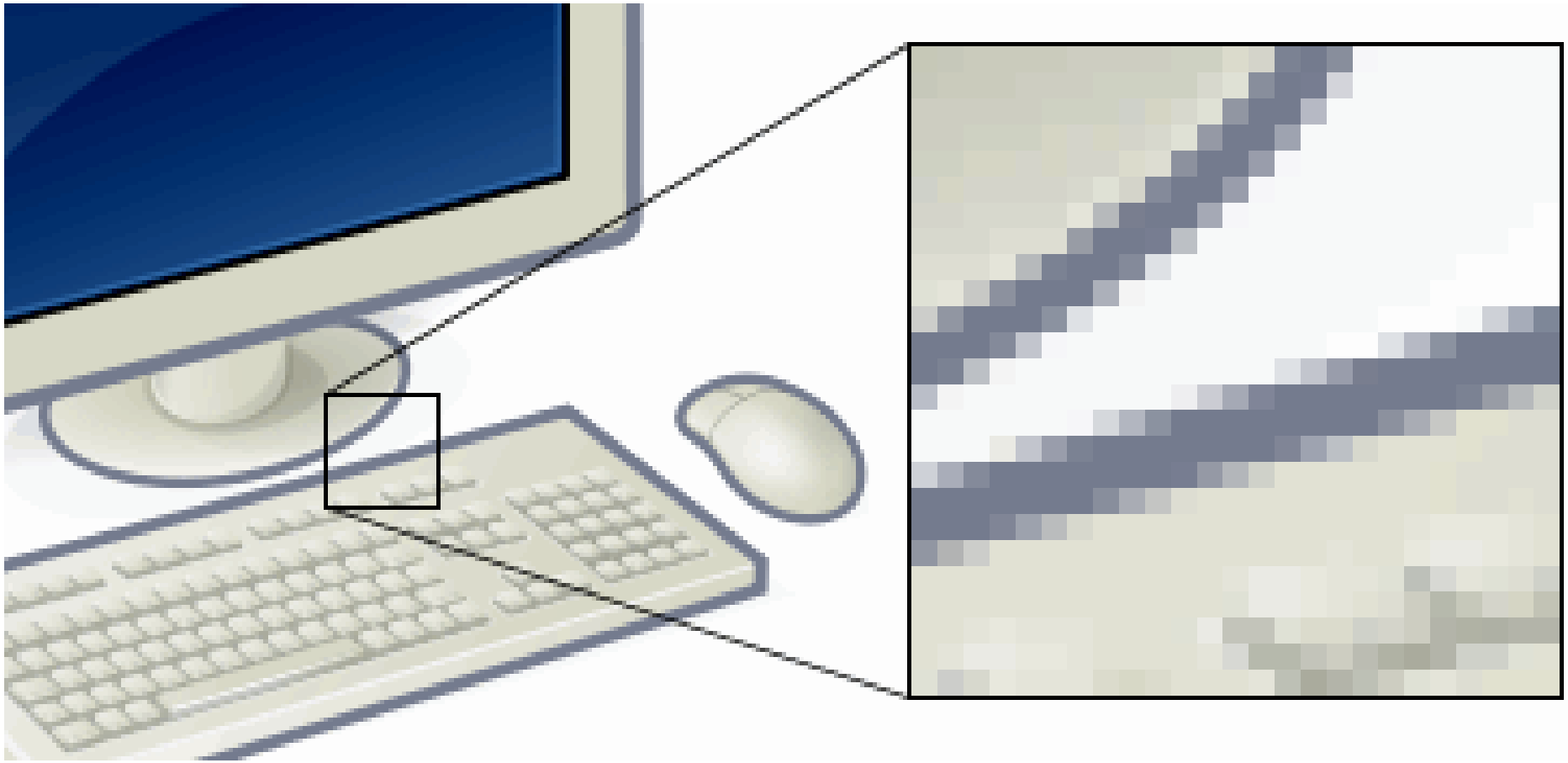


Image source: ed g2s, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pixel-example.png